HDOC Day-37

String Operations - Part 2

Question: Write a menu-driven C program using separate functions for the operations shown in the assignment (options 1-3 and 4 to exit).

C Program:

```
#include <stdio.h>

int length(char s[]) {
    int i = 0;
    while (s[i] != '\0') i++;
    return i;
}

void toUppercase(char s[]) {
    int i;
    for (i = 0; s[i] != '\0'; i++) {
        if (s[i] >= 'a' && s[i] <= 'z')
            printf("%c", s[i] - 32);
        else printf("%c", s[i]);
    }
    printf("\n");
}

void reverse(char s[]) {
    int i;
    for (i = length(s) - 1; i >= 0; i--)
        printf("%c", s[i]);
    printf("\n");
}

void palindrome(char s[]) {
    int i, n = length(s);
    for (i = 0; i < n / 2; i++) {
        if (s[i] != s[n - 1 - i]) {
            printf("Not a palindrome\n");
            return;
        }
    }
    printf("Palindrome\n");
}

int main() {
    char s[200];
    int choice, i;
    printf("Enter a string: ");
    fgets(s, sizeof(s), stdin);
    for (i = 0; s[i] != '\0'; i++) {
        if (s[i] == '\n') { s[i] = '\0'; break; }
    }
    do {
        printf("\n1. Convert to uppercase");
        printf("\n2. Reverse string");
        printf("\n3. Check palindrome");
        printf("\n4. Exit\nEnter choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1: toUppercase(s); break;
            case 2: reverse(s); break;
            case 3: palindrome(s); break;
            case 4: printf("Exit\n"); break;
```

```
            default: printf("Invalid choice\n");
        }
    } while (choice != 4);
    return 0;
}
```

Sample Output: For input madam: uppercase = MADAM; reverse = madam; palindrome = Yes. (Palindrome check is case-sensitive.)